Total No. of Questions : 8] [Total No. of Printed Pages : 4

Paper Code : 21306

F-406

B.C.A. (Second Semester) Examination, 2024

(New Course)

Paper No. BCA-N-201

DIGITAL ELECTRONICS

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. (a) Discuss various types of Logic Gates also discuss their applications.

   (b) Draw Circuit diagrams and truth table for following.

   (i) R-S flip flop

   (ii) D flip flop

   (iii) J-K flip flop

P.T.O.

2. (a) Using the K-Map method, simplify the following Boolean Function & obtain minimal SOP and minimal POS expression.

   $Y=\Sigma m(0,2,3,6,7)+\Sigma d(8,10,11,15)$

   (b) Discuss the difference between synchronous and asynchronous sequential circuit.

3. (a) What is the difference between multiplexer and demultiplexer? Explain in detail.

   (b) Explain excitation table, State table and state diagram.

4. (a) Show how a full adder can be converted to a full substractor with an addition of an inverter.

   (b) Explain the operations of master slave flip -flop and show how the race around condition is eliminated.

5. (a) Explain the various Boolean laws and simplify the following boolean expression.

(i) (A'+B+B'A)D

(ii) A'BC'+ABC'+A'B'C'+AB'C'

(iii) A'BC' +A'BC' D+BC'

(b) Give a detailed explaination using examples of conversion from binary to octal and octal to hexadecimal.

6. (a) Explain the difference between Encoder and Decoder. Draw circuit diagram for each.

(b) Define Multiplexer also draw circuit diagram for 8:1 multiplexer.

7. (a) Draw the internal diagram of a 4×1 multiplexer, clearly indicating the inputs and outputs. Explain the functionality using the function table.

(b) State and prove Associative law and distributive law with examples of each.

8. (a) Explain characteristic equation of JK flip flop from excitation table

(b) Design octal to binary encoder.